

"Physics as Information Processing" ~ Ander Aguirre ~ Discussion 5

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Hello and welcome everyone. It's September 30th, 2023, and we are in Physics as Information Processing. Discussion five should be a fun discussion on a great fifth lecture. So,

Ander, if you'd like to give some opening remarks, and then also anyone else, whether it's your first time joining a discussion or not, it would be great to hear your remarks on lecture five, or just where you're coming at it today. So, first though, Ander.

Sure. Hello, Daniel. So, I may as well start off by sharing the slides, right? What do you think?

Sure. So, just before, as a warm-up, let's review some of the things that Chris mentioned last time.

So, near the beginning, the main thing was this distinction between reductionist and scale-free theories. Obviously, one very salient feature of the free energy literature is that it falls under scale-free theories. These are theories where there is no preferred scale of energy or distance. reductionist theories are based on such a scale existing, right? You want to break the pieces into the fundamental building blocks of the Lagos. You know, people think of usually high energy or particle physics as the archetype of this thing. Whereas in scale-free theories, there is no such thing, and we should have the same skeletons or the same framework at different scales.

This is going to be very important. Chris, correct me if I'm wrong, but especially for the coming lecture where we talk about biology, Chris himself made a few remarks on how the renormalization group flow, aka the process of going from smaller to higher scale may be trivial in biology. That is to say that there is a self-similarity, okay? So, some features of biology should remain invariant at different scales. That is, what cells are to a body, different individuals of a species might be to an ecosystem, that sort of idea. So, after talking about scale, and perhaps when we get to the question parts, I guess this is the one thing that I would like Chris to explain a little more, more precisely what these diagrams mean. I have a rough idea, but it would be nice to hear him talk about that again. Anyways, in the second half, basically, we started asking, as is the theme in this course so far, from first principles, what do you need to talk about space and time? And there was some theorizing that time might be more fundamental. That is, the things that you need to implement a clock. We're thinking about what are the most basic constituents to implement a clock. And those may be even more basic than the things you need to implement QRFs or perception of space. After all, it seems like you need some spatial extension to perceive the space in the first place. I can't remember if it was somebody who was tuning in for the lecture, or a question afterwards, or something like that. But I think this remark gets to a very deep point. Somebody, I recall, mentioned that some organisms, it might have been Chris himself, I can't remember really, that there are organisms, you may know the name, Daniel, that are so tiny, that if they try to implement a gradient to try to chase their food or whatever, they're so small that they can't even perceive the gradient. In any attempt at implementing such perception, they're going to feel the same amount of concentration in all directions. So they're basically not large enough to perceive space. However, interestingly so, they can perceive time. I don't know how this ties to the remarks

that Chris made about space being more fundamental, I suppose. Chris was getting at the point that you need fewer parts to perceive space. And the fact that such small organisms can perceive time, but not space, I guess, is further empirical evidence for this thesis. And...

And actually, Andrew, if I can... Yes, please. I may not botch the explanation.

Yeah, I proposed earlier that time is actually more fundamental than space, that one has to have time to be able to see space. So that's further on in these slides, that issue is discussed.

Yeah. So do you want to say something about that, Chris?

Well, this is the right slide that...

Yeah, one just needs a memory, really, to have a clock, an external clock. I mean, to have an internal clock, when just one... One also has to have a memory to have an internal clock. But if you have a memory, a memory to have an internal clock, and there's some kind of activity that's periodic in the environment, then you have a clock, an external clock. But you don't need object identity to have a clock. And that's what distinguishes it from space. You have to have object identity, a clock, or in other words, motion, the ability to perceive motion to perceive space.

Right.

Yeah, any periodic behavior is effectively a clock.

Yeah.

Whereas, you know, for space, we need object persistence, which...

Two comments that we've made were that object persistence is the... Or rather, Chris made, is the defining feature of classicality. You said that last time, is that right, Chris? And also, that object persistence is precisely the notion of thickness, the way Friston would put it.

Yeah. Yeah. Object persistence is definitely a classical notion.

And it does seem to be the place where classicality enters quantum theory, as far as I can tell, is with this notion of object persistence. That that's the most fundamental place that it enters.

So how does that definition of... Or rather, relationship between classicality and quantumness consciousness being object persistence, how does it compare with other accounts coming from QDAR and decoherence and all of that?

Dr. Well, say quantum Darwinism is a good example. Quantum Darwinism assumes object persistence, right? It assumes that there is some persistent thing that interacts over time with the environment of... According to some interaction that has fixed eigenvalues. So...

So... Quantum Darwinism already assumes object persistence, and it also assumes classical communication, as we discussed a couple of times ago, and that it's a theory that is about the ability of multiple observers to agree about the state of some other object.

Mm-hmm.

Dr. Which, for them to agree about that object, not only does the object need to persist, the various observers need to have the same description of that object, or the same name for that object, or the same way of identifying that object. So yeah, quantum Darwinism actually assumes a lot of classical features.

The basic theory of environmental decoherence is, in effect, a single observer theory, so it doesn't have to assume classical communication, but it does assume object persistence.

Dr. Which, obviously, if you're going to talk about empirical science, you have to, right? You have to assume that your detector is a thing that persists through time, or you can't do an experiment. And more importantly than that, you have to assume that you're a thing that persists through time.

So... So we need these classical notions to talk about doing science.

Dr. Richard A.

I mean, I'm not the first person to point that out, of course. I mean, Bohr pointed that out in 1928, in his paper in Science, where he summarizes quantum theory. So it's a very, very old idea.

Dr. Right. And I guess, right away, I have a question that I suspect I may have asked before, but maybe we can revisit it from the perspective of today's discussion, which is... And maybe I should pull the slide. This is actually from session four. So if we're talking about differences between...

Dr. Richard A. Well, what the defining feature of classicality is, to me, that bears the question of...

Dr. Richard A. Here, right? This TQFT diagrammatic notation for the classical and the quantum channels.

So if we're talking about object persistence as the defining feature of classicality,

and this is the cartoon picture of what goes on on an EPR experiment,

Chris, could you say maybe a few words about how that classical... Like how the symmetry is broken between the quantum and classical channels? I suspect it might have been mentioned that these are symmetrical, but yet one of the two has to be a quantum and the other one is a classical channel.

The quantum channel mediates some sort of unitary evolution. Is that correct? Whereas the classical channel is like an information redundancy reservoir, which maybe has to do something with object persistence, but I'm very much speculating at this point. So I'm not sure if Chris, you have any comments?

Dr. Richard A. Yeah. Well, an easy way to think about this, I think, is to actually think about a Bell EPR experiment where you have two observers that are separated in space. So, you know, you have to assume here, though, to talk about Bell EPR, you have to assume the notion of space space to have a laboratory reference frame in which the two observers are spatially separated.

And in that situation, then the two observers both independently collect data for some amount of time on multiple replicates of some quantum state that's produced by a source. So you have pairs of photons or pairs

of electrons or pairs of whatever. And at some point, the data collection stops. And at that point, the observers

have to compare their results. So it's the comparison step that requires a classical channel.

And if you think about how this happens in practice, it happens via a file sent over to the internet, a paper that, or a piece of paper on which the data are written that's handed by one of the experimenters to the other experimenter. Or it happens by the experimenters talking to each other. So one experimenter encodes a

bunch of information in the ambient air, and the other experimenter hears it. Or it can be communicated by light signals, right? But it's always communicated by some medium. And that's what defines classical

communication. It's communication via a medium that persists through time in the observer's reference frame.

And the minimum time of the persistence of that medium is given by the speed of light. So if the observers are

a foot apart and they're communicating by a light signal, then it may only last a nanosecond.

But if they're talking to each other, you know, it lasts several seconds. And if they exchange a piece of paper, then that piece of paper may last for years in a lab notebook someplace. So

there's always the question of persistence over time

that is deeply bound up with this notion of classicality as a process that takes time.

information has to flow, has to be encoded in a medium and then read from that medium in some finite amount of time.

And in a sense, we're assuming something like that in the quantum description.

So when we think of this boundary, we think of this boundary via which the A and B,

the physical components interact, if we think of that boundary as a qubit array,

then each of those qubits persist through time. And that's what allows them to

to encode classical information. So encode observational outcomes that are repeated in time.

That's what allows them to serve as a memory, to encode information in a way that's thermodynamically irreversible. And if you think of the boundary in that way as a qubit array,

then it has a certain thickness in time. It's two Planck times thick.

The Planck time is 10^{-43} seconds. So it's unmeasurably small. But

the operation of interacting with a qubit is a physical interaction that,

from a relativistic point of view, has to take this minimal time.

So we're... It's in a sense, persistence for times that are significantly longer than the Planck time, that is the marker of classicality here.

But even these Planck times, even persistence over the Planck time,

is a necessary ingredient for talking about communication, even quantum communication.

So it's a very, very deep concept with a very long philosophical history, going back to, you know,

the ship of Theseus and all that. People have been wondering about,

what does it mean to say that something persists through time?

Okay. Does anyone want to make a comment on that or bring in a new question?

Please feel free to if you haven't brought up anything yet.

Sure. I can speak more. I mean, if you don't mind.

I also recall, and Chris, correct me if I'm wrong.

We said that related to what you just said,

that clock tick is always tantamount to one of these classical channels.

Is that correct? Did I remember that correctly?

Chris?

Yeah. An internal clock tick corresponds to an irreversible right on the boundary.

So one can then ask, what does it mean to say that that's irreversible?

Chris?

Chris What it means is it took some energy to do it.

And if you go back to read that bit again,
there's at least a reasonably high probability that it still
is in the same state that you wrote it in.

Chris Right.

Chris And clearly that probability is going to degrade through time,
because that action is on the boundary.

So whatever you've written, whatever bits that you've written are now exposed to the environment.
So if Alice writes them, they're now exposed to Bob and to Bob's dynamics.

Chris So as time goes on by your internal clock or any external clock,
the probability of that memory being the same,
when you go back to read it, slowly decreases, or maybe it decreases fast.

Chris And, you know, we know that from our own case, our own memories degrade through time.
Because the physical world is interacting with our brains in ways that we don't control.

Chris And, you know, killing neurons or changing their activity or whatever.

So whenever we talk about memory, it's always with this caveat that the memory is exposed to the
environment, and it's not guaranteed to be the same. It's only kind of

Chris Grace of the environment being reasonably benign that any of us had memories at all.

!

Carlos And so this thermodynamic cost to classical information sharing or writing,
is in a sense, would it be fair to say that, in a sense, the price you have to pay to protect against
decoherence or, you know, entangling this sector with the rest of the environment,
just to preserve its identity? Would that be fair to say?

Robert R. Yeah, I mean, you can think of decoherence as a good thing,
or you can think of decoherence as a bad thing, right? If you're trying to preserve
a quantum state, then it's a bad thing. You have to spend energy to prevent decoherence by,
you know, cooling everything down to milli-kelvins.

Chris Grace But if you're interested in preserving
classical data, then decoherence is a good thing, because it prevents your classical data from being
completely rearranged by becoming entangled with the environment. So, I mean, think about trying to
write classical memories using the states of atoms in the middle of the sun, and the temperature is so high
that you can't do anything. You can't rearrange the atoms in any way and expect them to stay put.
So, there you're in effectively a completely quantum environment,
just because of the high temperature. So, it's a plasma.

Robert R. Yeah, and then just one final question that popped to my mind as you say this,

and I can certainly leave it here for everyone else. So, if we're trying to protect against the random loss, against the loss of information, I suppose it really begs the question of what is the source of this noise that can perturb the bits in the boundary?

Robert R. Yeah, thank you,

picture of information uh processing is coming from the QRS failing to be aligned um then you know you hear about notions of quantum noise versus thermal noise uh Chris do you have any comments on all of that um yeah I think an important thing here

let me back up

yeah the notion of noise like so many other notions is very often uh treated as objective as observer independent and so when we're doing classical physics we think of noise as just you know there's there's some objective randomness in the universe and we call that noise so in this picture because we're working with just a bipartite decomposition right we just have Alice and Bob and that's it that's the entire universe um there's there's no external environment here to impose any sort of noise but since we only have two systems uh Alice and Bob equals everything else you know not Alice then uh if we talk about Bob being a noisy environment that's clearly relative to this boundary we've drawn between Alice and Bob so so noise isn't observer independent what's what's noise for Alice it's just Bob's dynamics right so so uh if if you think about where these systems come from you've got some universe that uh is in some pure state because it's an isolated system and uh that state is evolving evolving through time which means it's just undergoing phase rotation so there's no noise in that system right it's perfectly unitary etc etc as soon as you draw a boundary then there's information that the system on one side of the boundary doesn't have about the system on the other side so uh the individual systems are uh

uh

their states can now be considered conditionally independent i mean you can you can come up with a description in which their states are conditionally independent and that just means that what the other side is doing uh looks like noise unless uh Alice say can come up with a way of analyzing the data that she sees on

the data that she sees on her boundary uh in a way that makes it make sense right and then it's then it's then it's then it's we say it's information for Alice but what does that mean it just means it's behavior on her boundary that she's smart enough to assign some kind of semantics to oh and if she's not smart enough to do that

then it's just noise so as soon as you make the as soon as you make the notion of noise observer dependent it it turns

into something that makes a lot more sense

and and just to really finish this uh i guess it's been it's been a couple months at least but i think it was back in july that we

we talked about um the distinction between what you're saying which is uh things having the appearance of noise just because you

you you have finite resources so you cannot make sense of bob's internal dynamics as opposed to um in one

of in that cartoon of the

of the of the four possibilities for curf alignments right some of them were uh thermal in the sense that one one side was was acting on a on a boundary qubit while the other one was not so then it truly looks like like noise to the other side um does that does that get to the bottom of the distinction between quantum and thermal noise at all

Chris or am i misunderstanding things here yeah it yeah it turns the distinction into a essentially a relative a relative a q a qrf relative distinction right right if if uh you know uh i mean languages provide a good example i mean think of going to someplace where you don't understand the language at all like you know hapa new guinea or something like that i certainly wouldn't understand the language uh that that language language in that the behavior is speaking that language includes lots of information for those people and it codes no information at all for me so it looks like noise to me i can't make sense of it but it's not noise from their point of view um and now let's think about this need to have a thermodynamic sector if you're a

an organism that has to burn energy or any system that has to burn energy to to to act irreversibly on your environment uh that thermodynamic sector looks to you like noise right it's it's just temperature heat and uh you use that heat to drive your processing and if you're a really good heat engine then you can use it pretty efficiently and get you know a lot of bang for the buck in terms of information processing if you're a really inefficient heat engine then um maybe it you have to use almost your entire environment as environment as heat to do any little bit of information processing at all so that's not a statement about the universe as a whole that's the statement about you and your information processing capabilities and your and your thermal your heat processing capabilities you know how how good is your carno cycle determines uh the amount of information processing you can do thank you chris um ian

hi uh yeah thank you everyone for putting on this series it's the first time i've joined the discussion sessions and um yeah so it's a question about in one of the slides there was um a green rectangle and we were talking about that being a or chris was talking about that being a kind of um thinking of that as a bacterium um and it got me wondering whether i think how how else i could get an intuitive grasp of this kind of space-time emerging um and so what i was thinking is you know it appears like to us humans it appears like

the universe is expanding um through time and i was just trying to think you know a bacterium a bacterium sing if you placed a single bacterium on a petri dish um at time zero and proven something more simple you know let's say deep sea hydrothermal vent where life was supposed to are you know these um agents were supposed to have maybe begun existing as um phospholipid type molecules so i was just wondering you know if i was if i was a phospholipid within that bacterial membrane um just a single one single phospholipid and all i can measure is the existence of phospholipidness or not phospholipidness um or and selfness and not selfness then you know i was just trying to imagine as this bacteria divides after 20 minutes let's say the average sort of bacterial division time then before that happened all i could tell was i was surrounded by phospholipidness so there was no real distinction in my environment but once um the bacteria is divided

into two then i'm sitting and i'm observing some phospholipidness which isn't in my sort of what i call my own boundary um so then i've kind of created space space and time is starting to appear um even though the petri dish hasn't changed in size or my you know the universe hasn't really changed but it's the um phospholipids the orientation of the phospholipids so yeah i kind of i was just trying to liken it if this is a scale invariant principle how can we liken it to how other things might experience experience space and time emerging that's where i was coming from am i kind of totally barking up the wrong tree with that or is there is that no i i think that's i think that's a really good question and i i think you are you are constructing uh a very good kind of thought experiment and it's it's this sort of question that i'm really going to try to address in october of how how can we think about the fep as a scale theory uh in practice you know as as applying at the level of of macromolecules as you were just describing you know how do we come up with a description of what a phospholipid say can actually measure uh what actually affects its state and uh you know to what extent does it have a self-representation uh my guess is that a phospholipid probably doesn't but the bacterium seems to have at least some degree of self-representation i mean it has uh an immune system it can feel stress so it's starting to seem like uh an organism that's not that different from us uh but even even something like e coli it can cert it certainly behaves in what we call space but it's at least not clear to me that it has a representation of space uh although it it there are processes in the e coli that do distinguish say uh the two ends of the e coli i mean even if you think of the chemotaxis pathway uh the two ends of that pathway are quite distinct functionally so perhaps e coli has a one-dimensional representation of space it can tell its anterior end from its posterior end but perhaps it doesn't have any representation of what we would call left and right or up and down so um but if you look at multicellular organisms uh even something like say hydra uh is a spatially quite sophisticated kind of thing i mean it's not bilateral it's cylindrically symmetrical uh but it's got a very large distinction between anterior and posterior and it seems to know quite a bit uh at least i understand from people who work on hydra about um its own body the state of its own body and so i think these are exactly the same kinds of questions that we should be asking and they also apply at the scale that's much larger than us so if we think about uh a human society for example as an agent what can that agent measure and what can that agent represent and uh how complex can that agent's cognition be and i i think those questions are all essentially open so yeah it's a great thought experiment i think it addresses exactly the questions that we're interested in addressing here thank you chris harris and then anyone else with a raised hand yes thank you once again for allowing me to be here with all of you interesting as always so i have a question that is related to this notion of of the dual aspect of information basically that

the meaning of formation is not just a matter of interpretation
but rather it is rooted in the physical properties of the system under investigation
so basically that as i understand it means translates that the meaning of information
is not separate from the physical properties of the system under investigation
but it is rather an intrinsic part of it
so uh i i am also uh familiar um with this uh uh notion of uh uh from from the theory of computation
called the embrayments man's limit that is a limit on the maximum rate of computation that can be
achieved let's say in a self-contained system uh within the material universe
are written on the space so my question is how can we uh uh pass let's say
uh the though that that sort of uh uh computational uh limit in terms of uh information processing
I am not sure
that we can talk about a
I think it's very difficult to talk about
the classical information processing capability
of the universe as a whole
I mean one has to impose
some sort of classical description
and the question is then
where does that description come from
and from kind of whose point of view
is that a description
but certainly
as soon as we
you partition the universe
as a whole into
some system
and some other system
one can immediately talk about
their
information processing
capabilities
and then the question becomes
quantum information processing capabilities
versus classical information processing capabilities
and
if you're looking at classical information processing
to the extent that it's not reversible
it requires energy

and so one can immediately derive limits
based on the free energy access of that system
if you're talking about reversible information processing
which purely quantum information processing
is
then you don't have that energetic requirement
but you also don't have a well-defined answer
in any classical terms
in any classical terms
and
it's
it's only the
if
if Alice is your system of interest
that's able to do quantum information processing
then
it's only her interaction partner
Bob
who
can
can use her as a quantum computer
and then
his ability to
employ Alice as a quantum computer
is
limited by his
classical information processing
capabilities
because he has to define a classical input
and a classical output
so
I think this question becomes very complicated
as
as soon as it
is cast in these
observer relative terms
yeah
but

but
but such limits clearly exist
right
they
they just require certain assumptions
to define them
so
quantum information
can indeed be understood
as
a kind of dual to the classical information
right
um
yes
um
and
it's
it's essentially
one of these
bulk boundary type
dualities
that
that seem now to be ubiquitous
in physics
if we think of
um
I'm just thinking
in terms of the holographic principle
um
you can have
quantum processing
going on
in
in the bulk
in
in the system
that has a boundary
and

that can encode
it can
that can read classical information
from the boundary
and encode classical information
on the boundary
and
the
the
the

the

the
holographic principle
tells you how much
right
what can the maximum entropy
classical entropy
of the boundary b
um
but
what it doesn't tell you
is anything about the semantics
that
either the system inside
assigns to the information
or any observer outside
assigns to the information
it just
it's just a quantitative statement
so
in
in any setting
that

that's characterized
by that principle
um
the classical information
on the boundary
is
is dual
in an interesting sense
to the quantum processing
that's going on
inside
uh
in that
it
that classical information
on the boundary
encodes
everything
that an external observer
can know
about what's going on
on the inside
thank you
yes
thank you
yeah
okay
yeah
Anna
and then
Dean after
hi
thank you very much
um
I'm also
new to
um
your active inference

and your course

uh

I just want to say

that

uh

unfortunately

I work with humans

and

uh

um

but

and I say

unfortunately

because

uh

the complexities

are

horrific

um

so

uh

having said

that

maybe

my

my

my

question

will make sense

um

so

as I'm

learning

about

quantum

information

theory

and I'm

what I'm trying
to do
is assess
how can this
help me answer
questions that I have
in the systems
that I work with
uh
my question
is
in order to
have
a
time
arrow
irreversible
process
don't you need
to have
a goal
and
so
and so
my question
is
does it matter
to even
worry about
goals
because
as I
listen
to
all the
descriptions
and I think
about Alice

and Bob
and systems
I immediately
think about
well
so much
of what we're
talking about
depends upon
the goal
of the system
and
in some
very
abstract way
it seems
to me
that there's
a goal
that's
driving
the
need
for the
boundary
the need
for the
communication
um
and so
um
I'm
I
I'm so
obsessed
with this
question
that I

wanted to
ask
about
ask it
early on
before I
proceed
with my
learning
uh
yeah
that's
that's a
very good
question
you can
think of
the whole
free energy
principle
as
a
a
way of
of
describing
physical
systems
in a
way that
makes their
behavior as
agents
obvious
and
so the
the FEP
you could

state the
FEP by
saying that
every system
has a
goal
and
its goal
is
minimizing
prediction
error
right
it or
to put it
positively
its goal
is to
predict
and therefore
respond to
its
environment's
actions
in a
way that
preserves
its own
identity
and
I don't
know whether
you've read
much of
Carl
Fristin's
work but
he uses

the word
self
evidencing
a lot
it says
the goal
of any
system is
to
generate
evidence of
its own
existence
which sounds
very
psychological
and Carl
interestingly
enough
studied
first as
a
psychiatrist
so I
expect that
his point
of view
is actually
very consonant
with your
question and
where you're
coming from
here
yeah
actually
I started
I'm sorry

I didn't
make your
interrupt
no
no
go ahead
no
I started
I've
spent some
time
reading
um
Chris
Fristin's
um
ideas
and work
on
um
uh
the free
energy
principle
and
uh
self
evidencing
and yes
I will
continue
to dig
deep
uh
the problem
is
so I'm
an

anthropologist
and I've
worked
with
um
indigenous
people
and spent
20 years
of my
life
uh
living
in these
small
scale
societies
um
and
and so
one of my
observations
is that
and that
that I find
really
fascinating
um
but we
don't have
a good
understanding
of it
and that's
why
I decided
to
dig

into
physics
as
information
processing
uh
which is
that
goals
are
constantly
being
negotiated
and
it's
it's
uh
and more
along the
lines
of
so
Michael
Levine's
work
has
actually
was
what led
me to
explore
quantum
information
theory
um
which is
you know
like this

uh
collective
intelligences
that
are
exist
within
systems
and so
at each
level
of collective
intelligence
it's not
just that
you have
a collective
intelligence
dictating
a goal
but that
those goals
are being
negotiated
as
different
as the
environments
change
and as
relationships
between
the
systems
change
so
I know
these

these are
very
big
questions
but if
you could
just
like help
me
uh
give me
some
thoughts
on
what your
thoughts
have been
about
these
kinds
of
questions
um
yeah
I think
you're
pointing to
something
very
important
here
and
uh
uh
this
is
uh
a

question
that
um
this
kind
of
free
energy
principle
based
work
motivates
immediately
uh
when you
consider
a system
that
uh
comprises
a bunch
of
subsystems
um
you know
I
I work
with
Mike
Michael
Levin
at
Tufts
in
developmental
biology
yeah
and

one of
his
um
slogans
is
all
intelligence
is
collective
and
that
immediately
brings up
this issue
of
how do
these
components
as you
put it
uh
negotiate
their goals
how do
they
how do
they
uh
collaborate
how do
they compete
to what
extent
are they
trying to
deceive each
other
or coerce

each other
or all
of this
and
we're
you know
trying to
apply this
in a cell
biology
context
uh
as a
as a
description
of
development
and morphogenesis
if you think
about cells
in a
developing
organism
they're
constantly
facing
new
environments
uh
their
capabilities
are
constantly
changing
and
how they
interact
with each

other
is
constantly
changing
and this
all seems
very
psychological
sociological
so
I
I think
what you're
pointing to
is
the
ubiquitous
need
for this
kind
of
language
uh
in
areas
other than
the human
sciences
I mean
it's
language
that's
developed
within
the human
sciences
but it's
language

that seems
to be
important
um
at
much
smaller
scales
and
you know
clearly
one would
anticipate
it being
important
in much
larger
scales
you know
the scales
of
countries
or
civilizations
or
whatever
so
um
I guess
my answer
is yes
this is
an extremely
important
point of
view
uh
with

very
broad
implications
for how
we think
about
biology
at least
thank you
thank you

thank you
definitely
looking forward
to your
continued
learnings
and journey
thank you
so much
Dean
yeah my
question
has changed
three times
because
uh
just
the other
questions
that have
come up
in the
meantime
um
so
some context
here

Chris
you you
you've kind of
answered this
you mentioned
dual
as in
d-u-a-l
and then
Anna brought
up like
is there
some kind
of competition
to dual
a d-u-e-l
going on
and how do
we put
those two
things
together
and me
always thumping
the table
about minimum
two
we have to
have to keep
at least
two truisms
up at once
if we're going
to gain some
sense around
things
I want to go
back to that

lock
the LOCC
slide if we
can for a
second
because
Chris you
mentioned
your statement
was you draw
a boundary
and to me
there's a lot
of information
packed in
those four
words
um
you can
draw
and you
can undraw
you can
bound
you can
unbound
those are
perfectly
contextless
subject matter
information
carrying
frames
but you
also have
in that
packet
you

and
at
which
are
imperfect
orientations
an indefinite
article
and all
of us
occupying
a slide
here
none
of us
claiming
to be
anything
other
than
a
contradiction
so
one
aspect
of that
packet
two words
of it
literally
half
of it
is
perfectly
contextless
and is
information
carrying

and the
other
literally
half
of that
is
prediction
matter
or
information
graphing
it's
literally
the
byproduct
or
the
space
time
construct
that
is
included
so
I guess
what I'm
wondering
is
how do
we
make
these
different
matters
the
subject
matter
and

the
prediction
matter
tango
how do
we
get
them
to
relate
because
in the
session
five
you
went on
a detour
you said
we'll
take a
detour
which is
it holds
the exact
same
structure
as
you draw
a boundary
so there's
perfect
information
and there's
imperfect
information
in that
they're
dancing

we do
this all
the time
this seems
to me
at a
minimum
what
information
theory
is
and
that's
why
I
want
to
kind
of
go
back
to
this
slide
because
this
slide
was
in
a
much
earlier
session
but it
got to
session
five
where

the
context
could
be
recapitulated
and so
I'm
wondering
whether
or not
we can
really
talk
about
quantum
information
without
the
classical
or
if
that's
even
possible
well
I'll
go back
to
Boris
Boris
statement
in his
1928
paper
he
basically
said
the

answer
to
that
question
is
no
that
we
are
humans
we have
to
communicate
with
each
other
we
have
to
do
it
in
language
and
that's
classical
communication
and
if
while
while
we
are
doing
that
while
we're
having

this
conversation
we
can't
be
worried
about
the
question
of
do
I
still
exist
or
do
you
still
exist
we
can
worry
about
that
question
but
we
can't
do
it
at
the
same
time
that
we're
having
this

conversation
about
the
theory
of
lock
protocols
or
exchanging
data
from
a
bell
EPR
experiment
and
that's
of course
deeply
paradoxical
and
I
think
that's
just
the
situation
we're
in
you
said
none
of us
are
claiming
to be
anything
other

than
a
contradiction
it's
a good
thing
that we
are
self
conscious
about
the fact
that
we're
doing
something
contradictory
here
or
potentially
contradictory
here
a
colleague
and I
wrote
a
paper
several
years
ago
called
science
generates
limit
paradoxes
and
we

basically
rehearsed
a version
of this
very
discussion
that we're
having
right now
and
the
upshot
of that
paper
is that
in any
situation
in which
a
science
can
turn
its
theoretical
apparatus
onto
its
own
practice
so
for example
when
physics
can
turn
this
analysis
of

boundaries
onto
the
practice
physics
then it
generates
paradoxes
and
those
paradoxes
have
a
particular
structure
which
is
the
structure
that
priest
I can't
remember his
first name
a logician
who works
in
non
dialectic
logic
calls
a
limit
paradox
and
it
you know
all of

this is
coupled to
gardle's
theorem
and
these
other
paradoxes
of
self
reference
or
aspects
of
self
reference
but I
think this
situation
really is
ubiquitous
and
inescapable
that
as soon
as we
apply the
tools of
science to
the practice
of science
we
generate
these
paradoxical
situations
in which
we have

to acknowledge

that

we are

entities

of a

certain

sort

and

that

we

are

constructing

our

theories

using

a

faculty

of

imagination

that

lets us

act as

if we

don't

have a

point

of view

so it

lets us

act as

if

what we're

talking about

is point

of view

independent

but

our

theories
themselves
tell us
that we
have
points
of view
and that
they're
important
so we're
stuck in
this
paradoxical
situation
I don't
know whether
that's what
you're trying
to point
to here
but
it's
what
I immediately
leap to
hearing what
you said
yeah a
lot there
anyone else
want to
add a
piece
yeah I'm
gonna have
to go
away so

thank you
Chris
we'll
stay
for a
few
minutes
thank you
though
this was
awesome
we're
really
looking
forward
to
lecture
six
okay
thank you
all
and thank
you all
for attending
this
discussion
session
it's
nice
to have
this
kind
of
real
time
discussion
which is
missing

during
the
actual
lecture
sessions
so I
appreciate
it
thanks
again
see
you
thank
you
Chris
so
thank
you
Kate
do you
want to
ask
anything
or
anyone
can
can
just
bring
up
anything
they're
thinking
right
now
nothing
in
particular

I'm
just
really
looking
forward
to
the
last
lecture
on
the
biology
of
it
all
great
well
any
other
thoughts
or
questions
or
if
anyone
in
the
live
chat
has
a
thought
or
a
question
they
can
write

it
and
we'll
read
it
in
the
coming
minutes
well
I just
wanted
to thank
you
for
creating
the
space
it's
been
absolutely
amazing
that I
was able
to find
this
situation
where
people
are
talking
about
enormous
questions
very
difficult
questions
and

and
I
just
thought
I'd
share
that
I
find
in
academia
there's
I don't
know
why
we're
not
having
these
conversations
and
I've
been
it's
been
like
a
10
year
journey
and
I
have
finally
found
this
group
I

just
wanted
to
express
my
gratitude
thank
you
I'll
read a
question
from the
live chat
and anyone
can give
their
thoughts
so
mills
so
mills
8102
wrote
it
seems
that
consilience
would
aim
to
ameliorate
the
problems
of
self
reference
is
this

a
category
of
solution
that
seems
valuable
or
fraught
could
consilience
ameliorate
the problems
of self
reference
yes
dean
can I
just ask
a different
question
built on
that
which is
what if
the
contradiction
is a
feature
as opposed
to a
fraughtness
do I
contradict
myself
very well
then
I contain

multitudes
there's
the
Whitman
angle
then
there's
the
William
Blake
life is
a fiction
and is
made up
of
contradiction
so what
is the
problem
of
self
reference
whose
problem
is it
Harris
yeah
I will
try to
answer
a bit
to
tackle
a bit
both
questions
by
offering

my
views
from
the
cognitive
science
perspective
basically
I think
it has
to do
with
consilience
has to
do
with
numerous
cognitive
bias
that
we
all
are
let's
say
subjected
to
so
there
is
a
very
long
list
of
cognitive
biases
and

there
are
more
than
100
listed
I
think
so
this
is
one
of
the
challenges
in
terms
of
achieving
let's
say
good
levels
of
conciliant
across
various
domains
so
this
is
what
I
wanted
to
add
another
thought

on that
is
conciliant
though
it can
mean
many
things
synthesis
this
oh
sorry
oh
yeah
please
go
ahead
oh
yeah
I
just
so
I
don't
really
have
insights
into
the
self
referencing
problem
I need
to do
a lot
of
thinking
about

that
but
coming
from
training
in
evolutionary
biology
and
spending
you know
graduate
school
postgraduate
school
working
on
explanations
where
we
thought
of
many
hypotheses
models
etc
based
on
a goal
which is
organisms
are designed
to maximize
their fitness
and of
course
there are
lots of

you know
that there
are lots
of
problems
with that
and it
turns out
with our
data
what we
found
is that
was not
the case
in our
in the
populations
that we
working
with
too
often
the goal
did not
seem
to be
fitness
so
but
what
that
experience
gave me
is
that
there
was

something

we

could

hang

on

to

in

order

to

allow

ourselves

to

propose

models

for why

humans

behave

the way

they

do

and

the

decisions

that

they

make

so

that's

why

I

go

back

to

goals

!

because

if

whatever

level
of
collective
intelligence
we're
talking
about
there
is
a
goal
that
even
in
Harris
please
let me
know
if
this
would
be
correct
that
even
cognitive
biases
exist
they
exist
because
there
is
some
goal
and
therefore
we

can
talk
about
cognitive
biases
with
respect
to
some
goal
yes
there
is
some
sort
of
teleonomy
in
conscious
agents
or
personal
systems
as I
call
them
and
it
has
to
do
with
goals
yes
and
cognitive
biases
and

the
limitations
on
the
knowledge
let's
say
maybe
it's
limitations
on
parts
of
the
knowledge
representation
on
that
each
of
us
has
maybe
there
are
limitations
in
terms
of
access
to
knowing
and
there
are
various
forms
of

knowing
so
it
depends
on
many
things
and
for
example
one
of
the
cognitive
bias
that
I
believe
can
have
a
negative
impact
on
conciliant
let's
say
in
addressing
the
self
referentiality
will
be
the
halo
effect
so

where
you
seem
to
prioritize
let's
say
or give
some
advantage
to
some
types
of
individuals
ideas
or even
theories
because
of that
halo
or whatever
reason
that is
it is
based
on
maybe
on
many
different
things
maybe
on
your
past
history
or

other
things
so
yeah
thank
you
Anna
and
if
I
may
add
something
rather
briefly
your
question
reminded
me
of
Levin's
and I
think
it was
written
with
Phil
the
proposal
about
reduction
of
stress
as a
goal
that
guides
the

emergence
of
collective
structures
is
one
of
the
papers
formulated
that
that
seems
to
be
the
goal
that
drives
things
as
the
growing
complexity
reduction
of
collective
stress
or
reduction
of
stress
at
any
level
yeah
I don't
remember

the title
of the
paper
but
that
was
just
if I
understood
it
correctly
thank you
I'll look
that up
great
another
piece
that
of the
Levinson
Fields
collaboration
that
Chris
alluded
to
earlier
is
that
the
notion
of
computation
is
observer
dependent
that's
poly

computation

if

there

was

a

fully

encrypted

computation

an

observer

without

the

key

would

only

be

seeing

noise

to

them

whereas

if

they

were

to

have

the

decryption

key

then

they

could

fully

understand

and

go

into

more

detail
about
the
computation
so
then
let's
say
that
we're
in
relation
or
observing
or
investigating
some
life
phenomena
at
some
scale
but
we're
speaking
in the
scale
free
setting
but
we're
observing
a
thing
that's
there
people
have

ascribed
different
goals
to
describe
what
we call
goal
oriented
behavior
like
seemingly
to
survive
or
seemingly
to
have
evolutionary
fitness
however
one
an
observer
could
also
project
different
proximate
goals
like
the goal
to
minimize
stress
the goal
to
minimize

free
energy
the goal
to
minimize
surprise
those
proximate
observer
proposed
goals
as if
goals
may be
exactly
isomorphic
or aligned
with fitness
maximization
so
that could
just be
multiple
perspectives
of
projective
goals
and
whatever
the observer
projects
that
doesn't
mean
it
describes
the
actual

process
of
the
phenomena
at all
there's
just
nothing
Alice
can
make
any
story
about
Bob
and
it
may
or
may
not
have
anything
to do
or
even
be
possible
to be
of
the
kind
that
Bob
tells
about
himself
Dean

this
is
I
want
to
piggyback
on
that
Daniel
this
is
really
hugely
important
the
moment
you
use
the
word
goals
or
the
moment
you
use
the
word
fitness
you
are
assuming
context
whereas
a lot
of this
stuff
that's

happening
on this
holographic
screen
is
completely
abstractless
abstracted
and
completely
context
less
there's
no
no
context
required
so
if
you're
going
to
focus
on
that
part
of
it
as
long
as
you
can
juggle
both
at
the
same

time
and
keep
the
two
things
apart
because
one
does
not
require
context
the
other
completely
assumes
that
context
is
given
and
now
you're
taking
that
into
your
calculus
right
but
they're
not
the
same
thing
so
I

want
to
reinforce
what
Daniel
just
said
thank
you
Dean
Ian
thank
you
yeah
I was
just
maybe
reinforcing
what you
were saying
or
reflecting
on it
I was
thinking
about
the
behaviors
we see
in
other
animals
and
even
ourselves
that we
might
not

think
are
related
to
fitness
like
I don't
know
there's
those
kind
of
gazelle
type
creatures
that just
jump
into the
air
and like
but they
seem to
do that
at a
time
which might
make
them
be
predated
upon
by
some
tiger
or lion
which
you know
it's

you have
to
then all
the
sort
of
biologists
try and
figure out
how is
this
related
to
fitness
and we
may
do
the
same
you know
if I
go and
mow
my
lawn
waste
half a
Sunday
afternoon
mowing
my
lawn
what's
that
got
to
do
with

my
fitness
well
actually
it's
to do
with
keeping
up
with
the
joneses
and
my
social
status
so
it's
really
hard
to
actually
know
what
my
behaviour
is
about
I
can
create
a
story
to
say
that
it's
to

do
with
minimising
free
energy
or
fitness
but
really
is
it
or
not
I
don't
know
is
that
what
you
were
getting
at
or
something
different
yeah
that's
great
point
also
it's
come
up
several
times
about
goals

the
term
goal
is
not
to
be
found
within
the
active
inference
ontology
strictly
speaking
that's
why
we
have
so
much
fun
with
what
we
expect
and
what
we
prefer
how
can
it
be
what
we
expect
and

be
the
positive
thing
that
we
pursue
well
that's
how
paths
of
least
action
work
and
then
if
somebody
else
wants
to
look
at
that
path
of
least
action
and
say
well
here
was
a
mile
point
and

this
is
where
it
looked
like
they
were
heading
north
and
this
is
where
it
looked
like
the
goal
was
to
evade
this
those
are
all
stories
that
an
observer
could
tell
about
a
path
or
that
the

path
might
tell
itself
about
its
own
path
but
those
goal
narratives
don't
need
to
be
implemented
proximally
or
primally
with
goals
as
a
kind
of
core
player
in the
active
inference
ontology
I'll
read
some
of
these
great

comments
and
questions
just
if
anybody
has
a
thought
on
them
from
the
live
chat
just
feel
free
John
Reed
wrote
what
does
the
active
inference
group
I
think
that's
us
but
includes
everyone
think
about
the
equation

of
pilot
wave
with
many
worlds
recently
expounded
upon
by
Chris
O'Dowd
at
PBS
I
found
it
to
be
mixing
terms
and
frustrating
!
many
worlds
and
I
would
also
just
be
curious
on
the
physics
or
technical

side
where
does
what
Chris
is
sharing
connect
to
these
kinds
of
discussions
that
people
have
about
many
worlds
and
different
interpretations
of
quantum
yeah
that's
a great
question
so
I
don't
know
that
I
can
answer
it
but

I
know
that
when
it
comes
to
interpretations
of
quantum
mechanics
whether
you know
the
collapse
or
the
many
worlds
or
whatever
I
think
what
they're
hinting
at
is
and
Chris
would
probably
answer
something
along
those
lines
that

quantum
theory
is
fundamentally
about
information
and
until
we
make
sense
of
that
fully
and
fully
spell
out
all
the
details
I
think
these
philosophical
issues
are the
interpretations
of
quantum
mechanics
are going
to
linger
I
don't
have
any

strong
thoughts
on
interpretations
and
I
would
say
I'm
an
agnostic
about
interpretations
of quantum
mechanics
I
don't
know
that
I
subscribe
to
any
of
them
and
especially
Pilot
Wave
I
don't
have
the
slightest
idea
about
it
cool

yeah
Harris
yes
if
you
allow
me
I
have
a
small
story
regarding
Pilot
Wave
I
was
conversing
with
chat
GPT
and
I
actually
asked
what
it
thought
about
Pilot
Wave
after
watching
a
lecture
on
it
and

then
I
got
intrigued
and
I
wanted
to
ask
and
this
actually
happened
the
model
started
replying
and I
got
the
chance
to
read
a bit
the
reply
the
reply
was
like
it
is
very
interesting
but
most
probably
is

not
wrong
it
is
not
correct
it
is
wrong
and
then
it
must
have
reached
some
sort
of
information
goal
it
retracted
the
reply
and
it
replied
to
me
can
you
ask
me
a
different
question
so
this

is
what
this
is
my
short
experience
with
Pilot
Wave
here
ask
a
question
is
a
Pilot
Wave
more
or
less
than
100
feet
in
height
I
think
it's
exactly
exactly
100
any
other
thoughts
or
questions
it's

a fun
and
lively
live
chat
too
which
is
always
fun
yeah
well
we
are
coming
right
along
in
the
year
and
now
five
sixth
completed
with
this
course
so
any
comments
that
anyone
here
just
wants
to
make

in
a
closing
route
anybody
who
wants
to
can
go
for
it
yeah
I
have
a
question
if
questions
come up
have
come up
during
the
session
yes
how
can
send
when
could
I
send
them
to
and
how
great

question
in the
video
description
you can
get to
the syllabus
of the
course
and one
of the
pages
on the
course
website
course
syllabus
is a
question
and answer
form
so you'll
see
several
dozen
questions
that people
have already
written
and Chris
gives
answers
and then
it's
publicly
posted
all
of

the
lectures
discussions
and questions
and answers
will be
published
after the
course
so every
question
no matter
from where
or where
to
all of
these
questions
are encouraged
it only
makes it
a richer
and more
relevant
publication
thank you
okay
Ian
Harris
or Dean
yes
Susan
also
says
post
and
discord
great

point
yeah
I just
wanted to
add a
small
point
as a
conclusion
from my
personal
perspective
I really
find that
the ideas
of Chris
Fields
they can
be
revolutionized
revolutionary
let's
say
in a
variety
of
different
fields
and
I can
see it
in
quantum
cryptography
and also
about
what I've
mentioned

before
in terms
of
quantum
knowledge
representation
so
these are
the two
fields
that I
can
clearly
see it
so
it is
great
to be
able
to
participate
in
this
conversation
so
once again
thanks to
everyone
great to
participate
in this
conversation
or
quantum
knowledge
representation
whichever
you prefer

Andrew
any last
comments
not really
just looking
forward to
the next
one
I guess
and
next
few
I should
say
it
was
next
year
this
is
going
to
be
an
ongoing
thing
yes
we'll
come back
stronger
next
year
for
sure
okay
what a
ride
thank you

all

see you

next time

bye

bye

everyone

!

you